Sahaj Bhadja

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Summary

Backend Engineer with 2+ years shipping production **AI** features end to end in **TypeScript** and Python, from multi-provider LLM workflows and async job orchestration to APIs, databases, payments, and deployments used by real users.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant asynchronous **LLM orchestration** with background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent production load.
- Designed the multi-tenant data layer with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms, and shipped integrations through **CI/CD** at 99%+ reliability across 5K+ monthly jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- **Owned** an AI grading product end to end, from React UI and REST APIs to MySQL and AWS deployment, serving 150+ instructors and 10K+ daily requests in production.
- Built multi-step **prompt chains** across GPT-4o, Claude, and Qwen-VL with switchable providers, vision OCR fallbacks, and per-call cost tracking, reaching 90%+ grading consistency and cutting instructor feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Took data-sharing microservices from requirements to deployment with 20+ automated tests, and **led a team** redesigning ETL workflows on AWS Glue and S3 that boosted pipeline efficiency 68%.
- Optimized large-scale API integrations with pagination, field filtering, and caching, improving response times 30% for low-bandwidth users and cutting technical debt 20% through object-oriented refactoring of modules.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** Node.js and React platform with REST microservices, JWT authentication, and real-time WebSocket updates for billing and leads, syncing 2M+ records across multiple locations in real time.
- Shipped through Jenkins CI/CD with Docker and automated tests, and used product analytics on drop-off points to guide fixes that cut resource wastage 42% and raised **conversions** 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **payments** infrastructure for cards, wallets, and PayPal with idempotent webhooks for refunds and disputes, cutting manual handling about 70% across 50K+ booking records with multi-currency support.
- Built booking, pricing, and coupon APIs with a React Native mobile app and React TypeScript web dashboard, keeping responses under 200ms with **Firebase** real-time sync across Android, iOS, and web.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java
AI: LLM APIs, GPT-4o, Claude, Prompt Engineering, RAG, Embeddings, FAISS, Pinecone, OCR
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Microservices, JWT, Webhooks
Web & Mobile: React, Next.js, React Native, TailwindCSS, Stripe
Data & Infra: PostgreSQL, MySQL, MongoDB, Firebase, AWS, Vercel, Docker, Kubernetes, CI/CD, Git, Jest, PyTest

Projects

Prompt-to-App Generation Agent | *TypeScript, Node.js, LLM Tool Calling, PostgreSQL, Docker*

- Designed a **TypeScript** agent that turns an English app description into a plan, generates backend routes and database schema with tool calls, runs builds in a sandbox, and **self-repairs** failing tests.

Agent Run Evaluation Harness | *TypeScript, Node.js, PostgreSQL, LLM APIs*

- Designed an **eval** harness that replays saved user prompts against new agent versions, scores build success, latency, and token cost, and automatically blocks releases that **regress** output quality.

PiXMod Live | *Next.js, Stripe, SAM, Stable Diffusion, CLIP, MongoDB*

- Built and launched an AI **SaaS** product with SAM segmentation, Stable Diffusion editing, CLIP search, and Stripe subscriptions behind a Next.js interface, increasing premium conversion 18% for paying users.